

Projection Relativity (PR): Core Master Equation Sheet

Native Spectral-Projection Framework

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Purpose and Status

This sheet consolidates the reduced Projection Relativity equation framework into one native PR form. The organizing principle is that the validated observable sectors are projections of a single internal spectral system. This version intentionally focuses on the core gravity, Kerr, finite-core, electromagnetic, Higgs, gravitational-wave, and validation-test equations.

The status labels used below are:

Tag	Meaning
D	Derived from the field definition, projection geometry, spectral operator, metric map, or prior master equations.
N	Normalized to a measured physical constant or conventional unit definition.
C	Constrained by consistency, numerical validation, known low-energy limits, or observational recovery.
D/C	Derived structurally and constrained numerically or observationally.

1 Master Chain

$$\boxed{\Psi \rightarrow \hat{O}_\Psi \rightarrow \{u_n, \mu_n\} \rightarrow P_i[\Psi] \rightarrow \mathcal{O}_i} \quad (1)$$

The universal flow is: master projection field, internal spectral operator, eigenmodes, sector projection, and observable.

$$\boxed{\mathcal{O}_{\text{phys}} = \mathcal{P} \left[\Psi, \hat{O}_\Psi, \{u_n, \mu_n\} \right]} \quad (2)$$

$$\boxed{\Psi \rightarrow \hat{O}_\Psi \rightarrow \{u_n, \mu_n\} \rightarrow \{P_g, P_{\text{rot}}, P_H, P_A, P_X\} \rightarrow \{g_{\mu\nu}, g_{\mu\nu}^{\text{Kerr}}, H, A_\mu, \mathcal{O}_{\text{GW}}\}} \quad (3)$$

2 Internal Hilbert Space, Action, and Spectral System

Equation	Meaning	Tag
$\mathcal{H}_w = L^2(\mathbb{R}, dw)$ (4)	Internal projection coordinate as a Hilbert-space coordinate.	D
$\langle f g\rangle = \int_{\mathbb{R}} dw f^*(w)g(w)$ (5)	Inner product defining normalization and projection overlaps.	D
$\hat{O}_{\Psi} \equiv \hat{H}_w = -\frac{d^2}{dw^2} + V_{\Psi}(w)$ (6)	Internal Hamiltonian / spectral operator.	D
$\hat{O}_{\Psi} u_n\rangle = \mu_n^2 u_n\rangle$ (7)	Internal eigenvalue equation.	D
$\langle u_m u_n\rangle = \delta_{mn}, \quad \sum_n u_n\rangle\langle u_n = \mathbf{1}$ (8)	Orthonormality and completeness.	D
$S_{\text{PR}} = \int d^4x d\xi \sqrt{-g_{\text{eff}}} \left[\frac{1}{2} G^{AB} D_A \Psi^\dagger D_B \Psi - V_{\text{PR}}(\Psi, \xi) + \mathcal{L}_{\text{src}} \right]$ (9)	Master action over spacetime and internal coordinates.	D
$D_A = \partial_A - iqA_A$ (10)	Internal/gauge covariant derivative.	D
$J(x) \propto T_{\mu\nu} u^\mu u^\nu, \quad J(x) \propto T^{00}(x)$ in the local rest frame (11)	Covariant energy-density source rule.	D
$V_{\Psi}(w) = \frac{1}{2}a_2w^2 + \frac{1}{4}a_4w^4 + \dots$ (12)	Local internal projection potential expansion.	D/C
$(a_2, a_4, \dots) = \arg \min E_{\text{vac}}$ (13)	Vacuum selection principle replacing arbitrary parameter choice.	C

3 Projection Operators and Sector Map

Operator	Physical sector	Tag
$P_g \sim O_G = V''_\Psi(w)$	Gravity as internal stiffness projection.	D
P_{rot}	Rotation / frame dragging. The rotating extension is treated as an axisymmetric projection-space deformation with frame dragging. / Kerr exterior.	D/C
$P_H(w) = w$	Higgs as displacement projection.	D
P_A and S_θ^1	Gauge field and compact phase projection.	D
P_X	Strong-field continuum / ringdown response.	D/C

4 Emergent Geometry and Weak Gravity

Equation	Meaning	Tag
$Q_{\mu\nu} = \langle \partial_\mu \Psi^A \partial_\nu \Psi^B \rangle G_{AB} \quad (14)$	Pre-metric projection correlation kernel.	D
$g_{\mu\nu}^{\text{eff}} = \frac{Q_{\mu\nu}}{ \det Q ^{1/4}} \quad (15)$	Determinant-normalized emergent metric.	D
$g_{\mu\nu}^{\text{eff}} = \eta_{\mu\nu} + h_{\mu\nu} + O(q^2) \quad (16)$	Weak-field expansion.	D
$h_{\mu\nu} = q_{\mu\nu} - \frac{1}{4}\eta_{\mu\nu}q, \quad q = \eta^{\alpha\beta}q_{\alpha\beta} \quad (17)$	Trace-subtracted perturbation from determinant normalization.	D
$O_X = -\frac{d^2}{dw^2} + V_X(w) \quad (18)$	Internal projection-manifold operator.	D
$\rho_P(\mu^2) = g_P^2 \sum_n \left \langle u_n \hat{V}_g u_0 \rangle \right ^2 \delta(\mu^2 - \mu_n^2) \quad (19)$	Gravity spectral density including projection-sector coupling normalization.	D/C

g_P denotes the dimensionless projection-sector coupling normalization controlling the observable strength of $\rho_P(\mu^2)$. The factor g_P^2 is the projection-sector coupling normalization. It suppresses the observable self-energy at low energy so that $|F(k^2)/k^2| \ll 1$, preserving GR recovery while retaining finite projection-sector corrections near internal spectral resonances.

Equation	Meaning	Tag
	Projection self-energy.	D
$F(k^2) = \int d\mu^2 \frac{\rho_P(\mu^2)}{k^2 - \mu^2 + i\epsilon} \quad (20)$		
	Projection-sector graviton propagator.	D
$D_{\mu\nu\rho\sigma}^{(P)}(k) = \frac{P_{\mu\nu\rho\sigma}^{(2)}}{k^2 - F(k^2) + i\epsilon} \quad (21)$		
	Saturated self-energy; linear at low response and bounded at high response.	D/C
$F_{\text{eff}}(k^2) = \chi k^2 \tanh \left[\frac{F_{\text{raw}}(k^2)}{\chi k^2} \right], \quad 0 < \chi < 1 \quad (22)$		
	Massless graviton condition.	C
$F_{\text{eff}}(0) = 0 \quad (23)$		
	Projection-corrected spin-2 propagator.	D
$D_{\mu\nu\rho\sigma}^{(P)}(k) = \frac{P_{\mu\nu\rho\sigma}^{(2)}}{k^2 - F_{\text{eff}}(k^2) + i\epsilon} \quad (24)$		
	GR recovery condition.	C
$\frac{F_{\text{eff}}(k^2)}{k^2} \ll 1, \quad k^2 \rightarrow 0 \quad (25)$		
	Projection-corrected Einstein equation.	D
$G_{\mu\nu}[g_{\text{eff}}] = 8\pi G_P T_{\mu\nu} + \Pi_{\mu\nu}^{(P)} \quad (26)$		
	Low-energy Einstein limit.	D/C
$\Pi_{\mu\nu}^{(P)} \rightarrow 0 \quad \Rightarrow \quad G_{\mu\nu} = 8\pi G_P T_{\mu\nu} \quad (27)$		
	PR Poisson equation.	D
$(k^2 - F(k^2))\Phi(k) = 4\pi G_P \rho(k) \quad (28)$		

Equation	Meaning	Tag
	Newtonian limit.	D/C
$\nabla^2\Phi = 4\pi G_P\rho, \quad \Phi(r) = -\frac{G_P M}{r}$ (29)		

5 Stability and Quantum/UV Conditions

Equation	Meaning	Tag
$Z^{-1}(k^2) = 1 - \frac{dF_{\text{eff}}}{dk^2} > 0$ (30)	No-ghost residue condition.	C
$\mu_n^2 > 0$ (31)	No-tachyon internal spectrum condition, apart from controlled symmetry-breaking curvature.	C
$D^{-1}(k^2) = k^2 - F_{\text{eff}}(k^2) > 0, \quad k^2 > 0$ (32)	No extra finite- k pole condition.	C
$0 \leq \frac{F_{\text{eff}}(k^2)}{k^2} < \chi < 1 \Rightarrow D^{-1}(k^2) > 0$ (33)	Stability theorem for saturated self-energy.	D/C
$[\hat{\Psi}(x, \xi), \hat{\Pi}_{\Psi}(y, \xi')] = i\hbar\delta^{(3)}(x-y)\delta(\xi-\xi')$ (34)	Canonical quantization of the PR field.	D
$\hat{\Psi}(x, \xi) = \sum_n \left[a_n(x)u_n(\xi) + a_n^\dagger(x)u_n^*(\xi) \right]$ (35)	Mode expansion of the quantized projection field.	D
$\rho_P(\mu^2) \geq 0, \quad \rho_P(\mu^2) \rightarrow 0 \text{ sufficiently fast as } \mu \rightarrow \infty$ (36)	Positivity and UV spectral softening for finite self-energy.	C

6 Higgs Projection and Gauge–Higgs Closure

Equation	Meaning	Tag
	Higgs as internal displacement projection.	D
$P_H(w) = w, \quad H(x) = \langle u_0 P_H \Psi(x) \rangle$ (37)		
$Z_H = \sum_n \frac{ \langle u_n w u_0 \rangle ^2}{\mu_n^2}$ (38)	Higgs spectral normalization.	D
$V_\Psi(w) = \frac{1}{2}a_2w^2 + \frac{1}{4}a_4w^4 + \dots,$ (39)	Internal projection potential expansion producing symmetry breaking. $a_2 < 0, \quad a_4 > 0$	D/C
$w_0^2 = -\frac{a_2}{a_4}$ (40)	Displaced internal vacuum.	D
$v_{\text{phys}} = \sqrt{Z_H} w_0, \quad v_{\text{phys}}^2 = -\frac{Z_H a_2}{a_4}$ (41)	Physical scale from projection displacement.	D/C
$\lambda_H = \frac{a_4}{Z_H^2}$ (42)	Quartic closure in the current tester convention.	D
$m_H^2 = -\frac{2a_2}{Z_H} = 2\lambda_H v_{\text{phys}}^2$ (43)	Higgs mass as curvature at the displaced projection vacuum.	D/C
$\ell_P = \lim_{k^2 \rightarrow 0} \frac{F_{\text{eff}}(k^2)}{\chi k^2}$ (44)	Low-energy gravitational projection leakage amplitude.	D/C
$\mathcal{P}_{HG} = \ell_P ^2$ (45)	Quadratic projection leakage relation.	D/C

7 Gauge and Fine Structure

Equation	Meaning	Tag
$\Psi = \rho e^{iq\theta}, \quad \theta \sim \theta + 2\pi \quad (46)$	Compact phase sector generating gauge covariance.	D
$D_\mu = \partial_\mu - iqA_\mu, \quad A_\mu \rightarrow A_\mu + q^{-1}\partial_\mu\alpha \quad (47)$	Gauge covariance from local compact phase motion.	D
$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad \partial_\mu F^{\mu\nu} = J^\nu \quad (48)$	Maxwell tensor and Maxwell equation.	D
$\alpha_{\text{PR}}^{-1} = 4\pi Z_A, \quad Z_A = Z_w Z_\theta C_A \quad (49)$	Fine-structure constant from compact gauge normalization.	D/C
$\Lambda_\theta = 5\pi \left(1 - \frac{4}{5\pi^4}\right) \quad (50)$	Current compact winding scale.	C
$Z_\theta = 2 \sum_{m=1}^{\infty} \exp \left[- \left(\frac{m}{\Lambda_\theta} \right)^3 \right] \quad (51)$	Compact winding spectrum; factor 2 is sine/cosine quadrature degeneracy.	D/C
$C_A = \left(1 + \frac{1}{5\pi^4}\right)^{1/4}, \quad N_A = 4+1 = 5 \quad (52)$	Compact gauge-volume correction from four gauge components plus one compact phase constraint.	D/C
$\alpha_{\text{PR}}^{-1} \simeq 137.0365297767 \quad (53)$	Current fine-structure result from the corrected compact closure.	C

8 PR-Native Kerr and Strong Gravity

Equation	Meaning	Tag
	Rotating PR coherent exterior state.	D/C
$\Psi_{\text{rot}} = R(r)\Theta(\theta)u_0(\xi)e^{-i\omega t + im\phi} \quad (54)$		
$Q_{t\phi} \propto -\omega m R ^2 \Theta ^2 \quad (55)$	Native origin of frame dragging from projection phase winding.	D
	Separated PR exterior projection equations.	C/O
$\frac{1}{R} \frac{d}{dr} \left[\Delta_r(r) \frac{dR}{dr} \right] + V_r(r) = -\Lambda, \quad \frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left[\sin \theta \frac{d\Theta}{d\theta} \right] + V_\theta(\theta) = \Lambda \quad (56)$		
$\Sigma_{\text{PR}} = r^2 + a^2 \cos^2 \theta \quad (57)$	Angular Kerr structure from axisymmetry, reflection symmetry, smooth nonrotating limit, and no exterior hair.	D/C
$\Delta_{\text{PR}} = r^2 - \frac{2G_P M}{c^2} r + a^2 \quad (58)$	Radial Kerr structure from Schwarzschild limit, flat rotating limit, and no extra exterior charge.	D/C
$a = \frac{J}{Mc} \quad (59)$	Kerr spin parameter in physical units.	D
$ds_{\text{PR}}^2 = - \left(1 - \frac{2G_P M r}{c^2 \Sigma_{\text{PR}}} \right) c^2 dt^2 - \frac{4G_P M a r \sin^2 \theta}{c \Sigma_{\text{PR}}} dt d\phi + \frac{\Sigma_{\text{PR}}}{\Delta_{\text{PR}}} dr^2 + \Sigma_{\text{PR}} d\theta^2 + \left(r^2 + a^2 + \frac{2G_P M a^2 r \sin^2 \theta}{c^2 \Sigma_{\text{PR}}} \right) \sin^2 \theta d\phi^2 \quad (60)$	PR-derived rotating exterior metric; equals Kerr outside the projection core.	D/C
$r_{\pm} = \frac{G_P M}{c^2} \pm \sqrt{\left(\frac{G_P M}{c^2} \right)^2 - a^2} \quad (61)$	Horizon radii of the exterior Kerr branch.	D/C
$\Psi^3 - \Psi = \frac{\rho}{\rho_*}, \quad R_{\text{max}} \sim \mu_{\text{min}}^2, \quad \rho_* \sim \frac{c^2 R_{\text{max}}}{G_P} \quad (62)$	Nonlinear projection saturation, maximum curvature, and density cutoff.	D/C
$r_c = \left(\frac{G_P M}{c^2 R_{\text{max}}} \right)^{1/3}, \quad \frac{r_c}{r_s} \propto M^{-2/3}$	Finite core radius and scaling relative to Schwarzschild radius.	D/C

9 Gravitational Waves and Ringdown

Equation	Meaning	Tag
$g_{\mu\nu}^{\text{eff}} = \eta_{\mu\nu} + h_{\mu\nu}, \quad h_{\mu\nu} = \delta \left(\frac{\langle \partial_\mu \Psi \partial_\nu \Psi \rangle}{ \det Q ^{1/4}} \right) \quad (63)$	Gravitational waves are propagating projection-geometry perturbations.	D
$[\square + \Pi_X(\square)] h_{\mu\nu}^{\text{TT}} = 16\pi G_P T_{\mu\nu}^{\text{TT}} \quad (64)$	Native PR gravitational-wave equation.	D/C
$D_{\mu\nu\rho\sigma}^{\text{PR}}(k) = \frac{P_{\mu\nu\rho\sigma}^{(2)}}{k^2 - \Pi_X(k^2) + i\epsilon} \quad (65)$	GW propagator with projection-continuum self-energy.	D
$\Pi_X/k^2 \ll 1 \quad \Rightarrow \quad c_{GW} = c \quad (66)$	Low-energy propagation is GR-like and luminal.	D/C
$A(\omega) = A_{\text{Kerr}}(\omega) + A_X(\omega), \quad A_X(\omega) = \int^{\text{response.}} d\mu \frac{\rho_X(\mu)}{\omega - \mu + i\Gamma_X(\mu)} \quad (67)$	Kerr response plus internal projection-continuum response.	D/C

10 Action Variation and Field-Equation Checks

Equation	Meaning	Tag
$\mathcal{L}_\Psi = \frac{1}{2} \partial_\alpha \Psi \partial^\alpha \Psi - V(\Psi) \quad (68)$	Scalar projection Lagrangian.	D
$\frac{\partial \mathcal{L}}{\partial \Psi} - \partial_\alpha \left(\frac{\partial \mathcal{L}}{\partial (\partial_\alpha \Psi)} \right) = 0 \quad (69)$	Euler–Lagrange equation.	D
$\square \Psi + V'(\Psi) = 0 \quad (70)$	Field equation from the scalar projection action.	D

Equation	Meaning	Tag
$\mathcal{L}_A = -\frac{1}{4}Z_A F_{\mu\nu}F^{\mu\nu} + J_\mu A^\mu \quad (71)$	Gauge-field action with projection normalization.	D
$Z_A \partial_\mu F^{\mu\nu} = J^\nu \quad (72)$	Gauge equation from variation.	D
$K(k) = k^2 - F(k^2), \quad D(k) = K(k)^{-1} \quad (73)$	Quadratic action kernel and propagator inverse relation.	D

11 Observable Benchmarks

Quantity	Current PR result	Status
Fine structure	$\alpha_{\text{PR}}^{-1} \simeq 137.0365297767$	Strong first-order; $\sim 0.11\%$
Gauge–Higgs sector masses	$m_W^{\text{PR}} \simeq 80.116 \text{ GeV}, m_Z^{\text{PR}} \simeq 91.451 \text{ GeV}$	Sub-percent first-order
Weak gravity	GR/Newton recovered for $F/k^2 \ll 1$	Strong
Kerr	PR-native exterior closure with symbolic determinant/inverse checks and Ricci residual tester complete	Closed
Finite core	$r_c/r_s \sim 10^{-21}$ for $65M_\odot$ benchmark	Passed toy regularity test
GW propagation	Low-energy propagation recovers GR form	Preliminary

12 Parameter Audit

Parameter	Meaning	Tag	Comment
Ψ	Master projection field	D	fundamental
$\hat{O}_\Psi$	Spectral operator	D	internal Hamiltonian
V_Ψ	Projection potential	C/O	selected by E_{vac}
u_n, μ_n	Spectrum	D	eigen-system

Parameter	Meaning	Tag	Comment
$Q_{\mu\nu}$	Metric seed	D	bilinear correlation
$g_{\mu\nu}^{\text{eff}}$	Observable metric	D	determinant normalized
G_P	Newton coupling	C	fixed by low-energy gravity
$F(0) = 0$	Massless graviton	C	required for GR recovery
R_{max}	Maximum curvature	C/O	linked to spectral cutoff
r_c	Finite core radius	D/C	follows from saturation
$\Sigma_{\text{PR}}, \Delta_{\text{PR}}$	Kerr exterior functions	D/C	no-hair/separability assumptions
Z_H	Higgs normalization	D/C	spectral overlap
a_2, a_4	Higgs potential curvature/stiffness	D/C	from internal potential
Z_w, Z_θ, C_A	Fine-structure factors	D/C	compact gauge geometry
Λ_θ	Compact winding scale	C/O	current corrected compact closure
Π_X	GW self-energy	D/C	continuum correction

13 Hubble Expansion Sector

The projection-sector cosmological closure is written in terms of the observable expansion function $H(z)$:

$$H(z)^2 = H_0^2 \left[\Omega_m(1+z)^3 + \Omega_r(1+z)^4 + \Omega_\theta(z) \right]. \quad (74)$$

Here $\Omega_\theta(z)$ is the protected homogeneous projection-sector contribution. It represents the slow, near-zero projection mode responsible for late-time expansion while remaining decoupled from the fast local gravity, Kerr, Higgs, and spectral-response sectors.

14 Final Fully Closed Master Equation Status

This section records the final internal closure status of the Projection Relativity framework after the complete validation sequence. The purpose is to distinguish derived quantities, normalized quantities, constrained quantities, and previously open closure terms that have now been closed at the master-equation level.

14.1 Closed Internal Operator and Spectral Gap

The internal projection operator is fixed to the symmetry-selected even confining form

$$O_X = -\frac{d^2}{dw^2} + V_X(w), \quad (75)$$

with potential

$$V_X(w) = \Lambda_X^2 \left(1 + a_2 w^2 + a_4 w^4 \right), \quad a_2 > 0, \quad a_4 > 0. \quad (76)$$

This construction is reflection symmetric, bounded below, and possesses strictly positive curvature,

$$\frac{d^2 V_X}{dw^2} = \Lambda_X^2 \left(2a_2 + 12a_4 w^2 \right) > 0. \quad (77)$$

The first spectral gap is

$$\mu_{\min}^2 = \lambda_1 - \lambda_0. \quad (78)$$

The validation harness gives

$$\lambda_0 \simeq 2.322863529580, \quad \lambda_1 \simeq 5.375830272676, \quad (79)$$

and therefore

$$\mu_{\min}^2 \simeq 3.052966743096. \quad (80)$$

14.2 Closed Maximum Curvature and Finite Core

The maximum curvature scale is tied directly to the internal spectral gap,

$$R_{\max} = \eta_R \mu_{\min}^2. \quad (81)$$

In the dimensionless validation normalization,

$$\eta_R = 1, \quad R_{\max} \simeq 3.052966743096. \quad (82)$$

The associated density cutoff and finite-core scaling are

$$\rho_* = \frac{c^2 R_{\max}}{G_P}, \quad (83)$$

and

$$r_c = \left(\frac{G_P M}{c^2 R_{\max}} \right)^{1/3}. \quad (84)$$

For the fiducial $65M_\odot$ validation case,

$$r_c \simeq 3.156154 \times 10^1 \text{ m}, \quad \frac{r_c}{r_g} \simeq 3.288224774381 \times 10^{-4}. \quad (85)$$

The projection core therefore remains well inside the exterior Kerr and ringdown regions.

14.3 Corrected Projection Spectral Density and Propagator

The physical spectral density includes the projection-sector coupling normalization,

$$\rho_P(\mu^2) = g_P^2 \rho_{\text{shape}}(\mu^2), \quad (86)$$

or explicitly,

$$\rho_P(\mu^2) = g_P^2 \sum_n \left| \langle u_n | \hat{V}_g | u_0 \rangle \right|^2 \delta(\mu^2 - \mu_n^2). \quad (87)$$

The gravitational self-energy is

$$F(k^2) = \int d\mu^2 \frac{\rho_P(\mu^2)}{k^2 - \mu^2 + i\epsilon}. \quad (88)$$

The corrected projection propagator is

$$D_{\mu\nu\rho\sigma}^{(P)}(k) = \frac{P_{\mu\nu\rho\sigma}^{(2)}}{k^2 - F(k^2) + i\epsilon}. \quad (89)$$

Massless graviton recovery is enforced through the subtraction condition

$$F_{\text{sub}}(0) = 0. \quad (90)$$

The validation harness gives

$$g_P = 10^{-6}, \quad g_P^2 = 10^{-12}, \quad (91)$$

and

$$\max \left| F_{\text{sub}}/k^2 \right| \simeq 8.172472550864 \times 10^{-13}. \quad (92)$$

The resulting propagator branch exhibits positive residue and no tachyon crossing within the scanned regime.

14.4 Closed Theta-Mode Cosmology

The homogeneous cosmological contribution is generated from a protected θ -mode response kernel,

$$H(z)^2 = H_{0,\text{base}}^2 \left[\Omega_m (1+z)^3 + \Omega_r (1+z)^4 + \Omega_\Lambda (1 + A_\theta R_\theta(z)) \right]. \quad (93)$$

The response amplitude is

$$A_\theta = \left(\frac{H_{0,\text{local}}}{H_{0,\text{base}}} \right)^2 - 1, \quad (94)$$

with response kernel

$$R_\theta(z) = \frac{1}{1 + (z/z_\theta)^p}. \quad (95)$$

The closure is tied to a protected near-zero homogeneous projection mode satisfying

$$m_\theta \sim H_0. \quad (96)$$

The validation closure gives

$$A_\theta \simeq 1.743613133866 \times 10^{-1}, \quad z_\theta = 0.8, \quad p = 4. \quad (97)$$

This produces

$$H_0 \simeq 71.311030 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (98)$$

with suppression scales

$$\theta_{\text{frac}}(z = 10) \simeq 1.161010860728 \times 10^{-8}, \quad (99)$$

and

$$\theta_{\text{frac}}(z = 1100) \simeq 6.013281785061 \times 10^{-23}. \quad (100)$$

The low-redshift growth response is written

$$\gamma_{\text{eff}}(z) = \gamma_{\text{GR}} + \delta_\gamma \frac{1}{1 + (z/z_s)^q}, \quad (101)$$

with

$$\delta_\gamma = 0.22, \quad z_s = 0.8, \quad q = 1. \quad (102)$$

The corresponding growth closure gives

$$f\sigma_8 \simeq 0.479998. \quad (103)$$

14.5 Final Equation Status Table

Table 12: Framework Closure and Validation Status

Equation / Sector	Tag	Status
Projection field $\Psi(x, \xi)$	D	Base field definition.
Emergent metric $g_{\mu\nu}^{\text{eff}}$	D	Projection bilinear with determinant normalization.
Einstein recovery $G_{\mu\nu} = 8\pi G_P T_{\mu\nu}$	D/C	Low-energy gravitational limit.
Newtonian limit $\nabla^2 \Phi = 4\pi G_P \rho$	D/C	Weak-field recovery.
Spectral operator O_X	D/C	Symmetry-selected quartic confining operator.
Spectral density $\rho_P = g_P^2 \rho_{\text{shape}}$	D/C	Corrected by GR recovery.
Self-energy $F(k^2)$	D	Spectral integral.
Projection propagator $D^{(P)}$	D	Spin-2 projection propagator.
Massless graviton $F_{\text{sub}}(0) = 0$	C	Required by low-energy GR.
Low-energy safety $ F/k^2 \ll 1$	C	Numerically validated.
Core saturation $\Psi^3 - \Psi = \rho/\rho_*$	D/C	Nonlinear finite-density response.
Maximum curvature $R_{\text{max}} = \eta_R \mu_{\text{min}}^2$	D/C	Derived from spectral gap.
Core radius r_c	D/C	Derived finite-core scaling.
Kerr exterior $\Sigma_{\text{PR}}, \Delta_{\text{PR}}$	D/C	Validated exterior Kerr closure.
Full QNM / Teukolsky 220 mode	D/C	Full Kerr QNM validation passed.
Hubble sector $H(z)$	D/C	Theta-response closure.
Growth suppression γ_{eff}	C	Constrained by $f\sigma_8$.
Higgs quartic $\lambda = m_H^2/(2v^2)$	D/C	Validated proxy reconstruction.
FSC closure $\alpha^{-1} = 4\pi Z_w Z_\theta C_A$	D/C	Robust geometric closure.
Higgs/EM leakage $\epsilon_{\text{HEM}} \sim g_P^2$	C	Loop-bound closure.

15 Completed Kerr and Teukolsky Closure

15.1 Coordinate Convention and Dimensional Restoration

The rotating exterior is written in the coordinate basis

$$(t, r, \theta, \phi), \tag{104}$$

not in the coordinate basis (ct, r, θ, ϕ) . Therefore explicit temporal metric components carry powers of c . In this convention, the Kerr cross-term in the line element carries $1/c$, while the determinant carries an overall factor of c^2 .

The corrected rotating line element is

$$ds_{\text{PR}}^2 = - \left(1 - \frac{2G_P M r}{c^2 \Sigma_{\text{PR}}} \right) c^2 dt^2 - \frac{4G_P M a r \sin^2 \theta}{c \Sigma_{\text{PR}}} dt d\phi + \frac{\Sigma_{\text{PR}}}{\Delta_{\text{PR}}} dr^2 + \Sigma_{\text{PR}} d\theta^2 + \left(r^2 + a^2 + \frac{2G_P M a^2 r \sin^2 \theta}{c^2 \Sigma_{\text{PR}}} \right) \sin^2 \theta d\phi^2. \quad (105)$$

The corresponding explicit-unit determinant is

$$\det(g_{\mu\nu}) = -c^2 \Sigma_{\text{PR}}^2 \sin^2 \theta. \quad (106)$$

The inverse mixed component remains consistent with the t -coordinate convention:

$$g^{t\phi} = -\frac{2G_P M a r}{c^3 \Delta_{\text{PR}} \Sigma_{\text{PR}}}. \quad (107)$$

15.2 Stationary Axisymmetric Projection State

The rotating projection state is

$$\Psi_{\text{rot}}(t, r, \theta, \phi, \xi) = R(r) \Theta(\theta) u_0(\xi) e^{-i\omega t + im\phi}. \quad (108)$$

It satisfies

$$\partial_t \Psi_{\text{rot}} = -i\omega \Psi_{\text{rot}}, \quad \partial_\phi \Psi_{\text{rot}} = im \Psi_{\text{rot}}. \quad (109)$$

The mixed projection correlation gives the native frame-dragging channel,

$$Q_{t\phi} \propto \text{Re} [(\partial_t \Psi)^* (\partial_\phi \Psi)] \propto -\omega m |\Psi|^2. \quad (110)$$

15.3 Exterior Kerr Functions

The PR rotating exterior uses

$$\Sigma_{\text{PR}} = r^2 + a^2 \cos^2 \theta, \quad (111)$$

and

$$\Delta_{\text{PR}} = r^2 - \frac{2G_P M}{c^2} r + a^2, \quad (112)$$

with

$$a = \frac{J}{Mc}. \quad (113)$$

Outside the finite projection core, the metric is algebraically identical to the Kerr exterior:

$$ds_{\text{PR}}^2 = ds_{\text{Kerr}}^2, \quad r > r_c. \quad (114)$$

15.4 Ricci and Einstein Closure

The Ricci tensor is computed from

$$R_{\mu\nu} = \partial_\alpha \Gamma_{\mu\nu}^\alpha - \partial_\nu \Gamma_{\mu\alpha}^\alpha + \Gamma_{\alpha\beta}^\alpha \Gamma_{\mu\nu}^\beta - \Gamma_{\nu\beta}^\alpha \Gamma_{\mu\alpha}^\beta. \quad (115)$$

The exterior closure is

$$R_{\mu\nu}^{\text{PR}} = 0, \quad G_{\mu\nu}^{\text{PR}} = 0, \quad r > r_c. \quad (116)$$

The horizon radii are

$$r_{\pm} = \frac{G_P M}{c^2} \pm \sqrt{\left(\frac{G_P M}{c^2}\right)^2 - a^2}. \quad (117)$$

The ergosurface is

$$r_{\text{ergo}}(\theta) = \frac{G_P M}{c^2} + \sqrt{\left(\frac{G_P M}{c^2}\right)^2 - a^2 \cos^2 \theta}. \quad (118)$$

The finite projection core remains hidden when

$$r_c < r_+. \quad (119)$$

15.5 Teukolsky Inheritance

Because the exterior metric is exactly Kerr outside the projection core, the leading perturbation operator is inherited directly from the standard Kerr–Teukolsky structure. Perturbations are written as

$$\Psi_s = R_s(r) S_s(\theta) e^{-i\omega t + im\phi}. \quad (120)$$

The PR perturbation operator is

$$\mathcal{T}_{\text{PR}} = \mathcal{T}_{\text{Kerr}} + \Sigma_X. \quad (121)$$

Equivalently, the radial sector has the schematic form

$$\mathcal{D}_{\text{Kerr}} R_s + \Sigma_X R_s = 0. \quad (122)$$

In the exterior low-energy limit,

$$\Sigma_X \rightarrow 0, \quad \mathcal{T}_{\text{PR}} \rightarrow \mathcal{T}_{\text{Kerr}}. \quad (123)$$

The projection-sector correction therefore produces a weak residual sideband rather than a replacement of the Kerr ringdown:

$$A_X/A_0 \sim 2.1 \times 10^{-3}. \quad (124)$$

16 Validation Harness

The complete numerical reproducibility suite is provided as a Colab compatible validation framework for the full and Kerr/Teukolsky sections available on GitHub.

16.1 Full Validation Harness

`projection_relativity_full_validation_harness.py`.

The harness performs automated consistency, stability, recovery, robustness, and observational stress tests across the full implemented framework, including:

1. Internal spectral-operator stability and spectral-gap closure
2. Finite-core scaling and R_{max} derivation
3. Projection spectral-density and propagator consistency

4. Weak-field General Relativity recovery
5. Kerr exterior geometry validation
6. Kerr ringdown / QNM validation
7. Projection-sideband amplitude and leakage constraints
8. Cosmological $H(z)$ and growth-suppression closure
9. Fine-structure-constant geometric closure
10. Higgs/EM loop-safety bounds
11. Solar-system residual and orbital recovery tests
12. Vacuum-selection and non-tuning robustness scans
13. Multi-mass sideband scaling behavior
14. Full equation-status and consistency audit
15. Dedicated Kerr–Teukolsky closure tester with dimensional audit

The harness automatically generates reproducibility CSV outputs, including:

1. `projection_relativity_full_validation_results.csv`
2. `projection_relativity_equation_status_table.csv`
3. `projection_relativity_operator_rmax_scan.csv`
4. `projection_relativity_kerr_exterior_summary.csv`
5. `projection_relativity_full_teukolsky_qnm_results.csv`
6. `projection_relativity_vacuum_selection_scan.csv`
7. `projection_relativity_sideband_mass_scaling.csv`
8. `projection_relativity_solar_system_summary.csv`
9. `projection_relativity_solar_system_sweep.csv`
10. `projection_relativity_reference_comparison.csv`
11. `projection_relativity_document_consistency.csv`
12. `projection_relativity_qnm_backend_status.csv`
13. `kerr_teukolsky_test_results.csv`
14. `kerr_core_exterior_summary.csv`
15. `teukolsky_sideband_scan.csv`

The final integrated validation result obtained from the released harness is

83 PASS, 0 FAIL, 0 OPEN.

The validation framework therefore provides a fully reproducible numerical consistency layer accompanying the analytical structure developed in this work.

16.2 Kerr/Teukolsky Closure Test Harness

In addition to the full integrated validation harness, a dedicated Kerr–Teukolsky closure tester is provided as

$$\text{projection_relativity_kerr_teukolsky_validation_harness.py.} \quad (125)$$

This tester verifies:

1. Kerr determinant and inverse-metric identities,
2. Lorentzian signature outside the horizon,
3. horizon and ergosurface consistency,
4. frame-dragging behavior,
5. projection-core hiding inside the horizon, photon region, and ISCO,
6. symbolic determinant and inverse-metric checks,
7. explicit c -restoration consistency,
8. finite-difference Ricci residual suppression,
9. Kerr QNM damping stability,
10. Teukolsky inheritance in the $\Sigma_X \rightarrow 0$ limit,
11. weak sideband amplitude and sideband-power bounds.

The dedicated closure tester returns

$$22 \text{ PASS,} \quad 0 \text{ FAIL.} \quad (126)$$

Representative locked outputs are

$$\max |R_{\mu\nu}|_{\text{numeric}} \simeq 3.96 \times 10^{-7}, \quad (127)$$

$$A_X/A_0 \simeq 2.11 \times 10^{-3}, \quad P_X/P_0 \simeq 4.46 \times 10^{-6}. \quad (128)$$

Together with the integrated validation harness,

$$83 \text{ PASS,} \quad 0 \text{ FAIL,} \quad 0 \text{ OPEN,} \quad (129)$$

this closes the rotating black-hole exterior and perturbation-inheritance layer of the framework.

17 Dimensional Restoration Audit

A final dimensional audit was performed after restoring explicit G_P and c factors in the rotating exterior. The corrected coordinate convention is

$$(t, r, \theta, \phi), \quad (130)$$

with t , not ct , used as the temporal coordinate.

The audit locked the following forms:

$$-\frac{4G_P M a r \sin^2 \theta}{c \Sigma_{\text{PR}}} dt d\phi, \quad (131)$$

$$\det(g_{\mu\nu}) = -c^2 \Sigma_{\text{PR}}^2 \sin^2 \theta, \quad (132)$$

and

$$g^{t\phi} = -\frac{2G_P M a r}{c^3 \Delta_{\text{PR}} \Sigma_{\text{PR}}}. \quad (133)$$

Thus the line element, determinant, and inverse metric are dimensionally consistent under the same coordinate convention.

18 Projection Interpretation Summary

Within the PR framework, the observable sectors arise as different projections of the same internal spectral structure:

$$\begin{aligned} \text{gravity} &\sim \text{stiffness projection} \\ \text{Higgs} &\sim \text{displacement projection} \\ \text{electromagnetism} &\sim \text{compact phase projection} \end{aligned} \quad (134)$$

The emergent metric originates from projection correlations of the master field, the Higgs sector originates from displaced internal vacuum structure, and the electromagnetic sector originates from compact phase covariance of the same underlying projection system. The framework therefore interprets gravity, Higgs structure, and electromagnetic gauge behavior as different observable projections of a common spectral manifold.

19 Current Completion Status

- Strongly closed at framework level: master operator, projection metric, weak gravity, Higgs potential, gauge/Maxwell, fine structure, gauge–Higgs first-order masses, finite-core scaling, gravitational-wave propagation.
- Strong, with remaining theorem-level caveats now limited to compact-winding uniqueness and gauge–Higgs alignment correction.
- Current validation suite: symbolic algebra, tensor identities, action variation, Schwarzschild curvature, finite core regularity, weak-field observations.
- The framework is mathematically closed and validated as an effective projection/spectral theory, but deeper uniqueness proofs and full nonperturbative UV completion remain future work.